

2.1 Decomposition model

ESTP Training
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eurostat 

Regression model

- *Additive case:*

$$y_c \equiv y - \bar{\delta} \cdot reg_{fixed}$$

$$y_c = \alpha \cdot cal + \beta \cdot out + \gamma \cdot reg + \mu$$

- *Multiplicative case:*

$$y_c \equiv \ln(y) - \bar{\delta} \cdot reg_{fixed}[-lp]$$

$$\ln(y_c) = \alpha \cdot cal + \beta \cdot out + \gamma \cdot reg + \mu$$

- $cal = [tde \quad ee \quad omhe]$
- $out = [out_t \quad out_s \quad out_i]$
- $reg = [reg_t \quad reg_s \quad reg_i \quad reg_{sa} \quad reg_y \quad reg_u]$
- $\mu = \text{stochastic component}$

Handling of the regression variables

Code	Description	T	S	I	SA
tde	Trading days (default, holidays, user-defined)		x		
ee	Easter		x		
omhe	Other moving holidays (TODO: Ramadan...)		x		
AO	Additive outlier			x	x
TC	Transitory change			x	x
LS	Level shift	x			x
SO, SLS	Seasonal outlier / seasonal level shift		x		
Reg_i	Regression variables allocated to irregular, IV (p)			x	x
Reg_t	Regression variables allocated to trend, ramps, IV (p)	x			x
Reg_s	Regression variables allocated to seasonal, IV (p)				
Reg_sa	Regression variables allocated to seas. adjusted				x
Reg_y	Regression variables removed before decomposition				
Reg_u	Regression variables unallocated to a component	p	p	p	p

p stands for partially

Decomposition

- Additive case:

$$\gamma_u \cdot reg_u + \mu = y_{cmp} = t_{cmp} + s_{cmp} + i_{cmp}$$

- Multiplicative case:

$$\gamma_u \cdot reg_u + \mu = y_{lin} = t_{lin} + s_{lin} + i_{lin}$$

$$\exp(\gamma_u \cdot reg_u + \mu) = y_{cmp} = t_{cmp} \cdot s_{cmp} \cdot i_{cmp}$$

Final decomposition

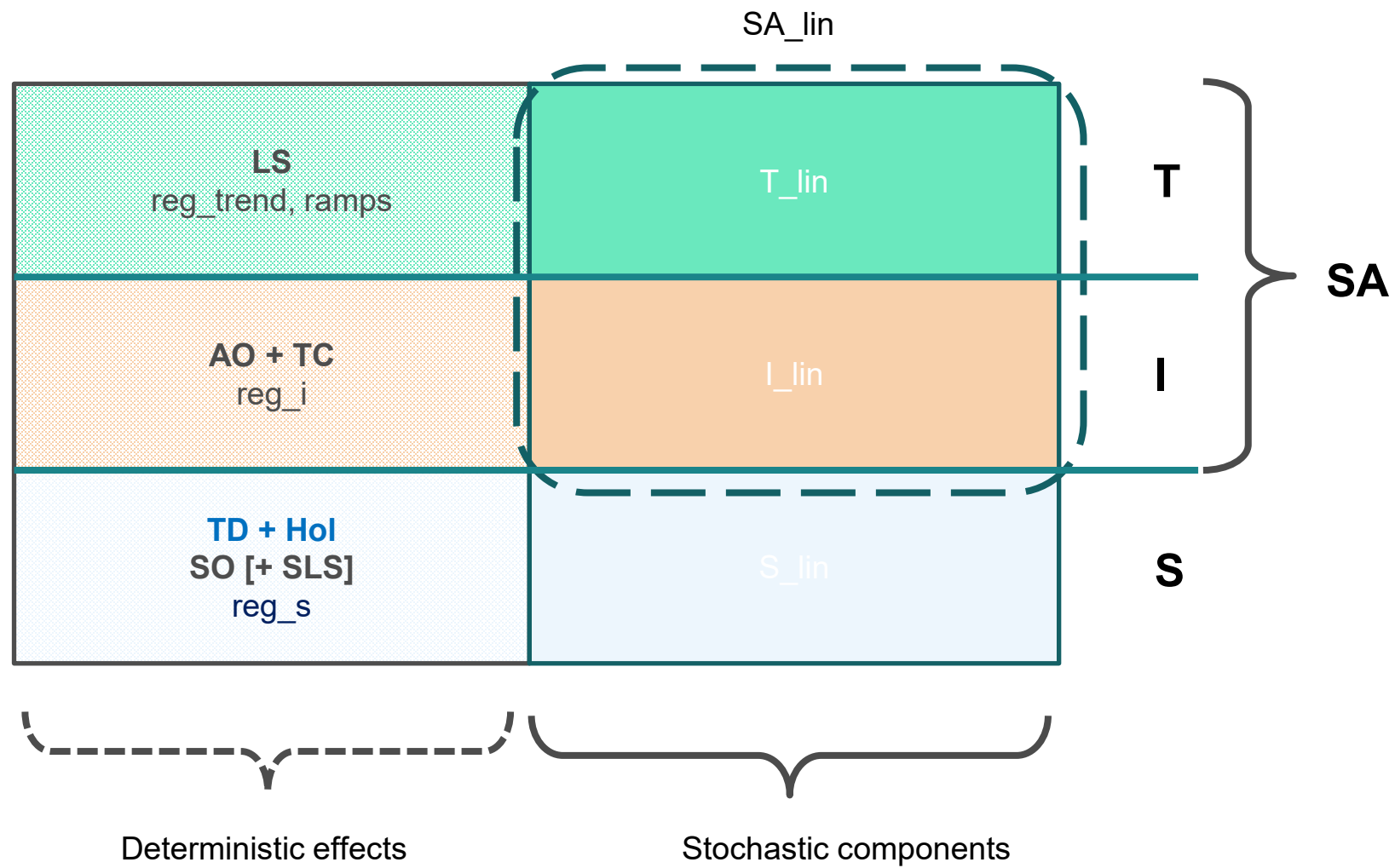
- **Additive case:**

- ✓ $t = Out_t + Reg_t + t_{cmp}$
- ✓ $s = Cal + Out_s + Reg_s + s_{cmp}$
- ✓ $i = Out_i + Reg_i + i_{cmp}$
- ✓ $sa = t + i + Reg_{sa} = y_c - Reg_y - s$
- ✓ $y_c = s + t + i + Reg_y$

- **Multiplicative case:**

- ✓ $t = \exp(Out_t \cdot Reg_t) \cdot t_{cmp}$
- ✓ $s = \exp(lp \cdot Cal \cdot Out_s \cdot Reg_s) \cdot s_{cmp} = \exp(\widetilde{Cal} \cdot Out_s \cdot Reg_s) \cdot s_{cmp}$
- ✓ $i = \exp(Out_i \cdot Reg_i) \cdot i_{cmp}$
- ✓ $sa = t \cdot i \cdot \exp(Reg_{sa}) = (y_c / \exp(Reg_y)) / s$
- ✓ $y_c / \exp(Reg_y) = s \cdot t \cdot i$

$$Var_x = \gamma_x var_x + \overline{\delta_x} var_{x_{fixed}}, \quad var = out, reg, cal$$



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